

Phonetics: The Articulation of Speech Sounds

-  *The Speech Mechanism*
-  *The Respiratory System*
-  *The Articulatory System*
-  *The Phonatory System – The Vocal Cords*
- Voiceless and voiced sounds*
- The vocal cords and pitch*

We learnt in Chapter 3 that **Phonetics** is a component of linguistics and that it deals with the **articulation**, **transmission** and **reception** of speech sounds. In this Chapter, we shall discuss the **ARTICULATION** of speech sounds in some detail.

It may come to you as a matter of surprise, but about half of the human body is needed for the production of speech. The organs of our body that help us to speak can be divided into three groups. Each group of human organs that help people to speak forms a **system**. The way in which these three systems help us to speak is called the **SPEECH MECHANISM**. These three systems are:

- (a) The **RESPIRATORY** system;
- (b) The **ARTICULATORY** system; and
- (c) The **PHONATORY** system.

The **RESPIRATORY** system lies in the **TRUNK** of our body (that is, from the **NECK** to the **ABDOMEN**). The **ARTICULATORY** system lies in the **HEAD** and the **PHONATORY** system lies in the **THROAT**. Let us look at these three systems in some detail.

(A) THE RESPIRATORY SYSTEM

The respiratory system consists of:

- (i) the **LUNGS**;
- (ii) the **MUSCLES OF THE CHEST** which make the lungs expand and contract;
- (iii) the **WINDPIPE** (which is also called the **TRACHEA** /trə'ki:ə/);
- (iv) the two tubes which enter the lungs. These two tubes are called **BRONCHI** /'brɒŋkai/. Each tube is called a **BRONCHUS** /'brɒŋkəs/.

The **lungs** are soft bodies situated on either side of our chest. The lungs receive oxygen through the two **bronchi**. The two bronchi join together into one **bronchus** above the lungs. The bronchus joins the **windpipe** or the **trachea**.

When we breathe **in** (that is, when we take the air from the outer atmosphere into our lungs), we take the air into the lungs through the windpipe, the bronchus and the bronchi. The process of **breathing in** is called **INSPIRATION**. When we take air into our lungs, the muscles of

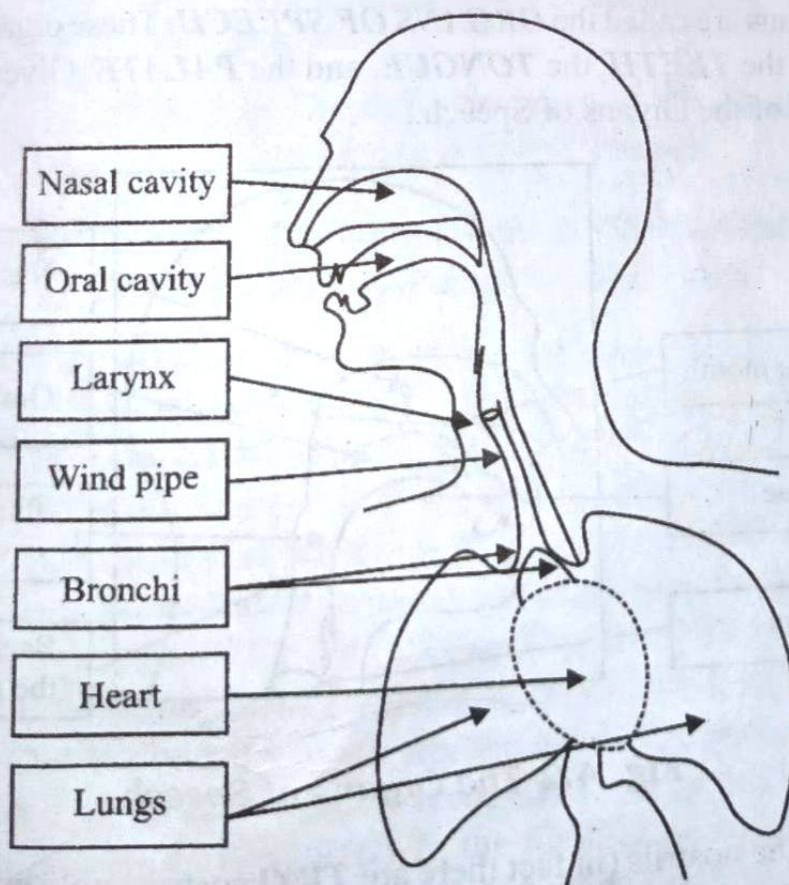


Fig. 4.1 The respiratory System and the Heart

the chest make the lungs *expand*. When we breathe *out*, the air from the lungs goes out of the lungs through the bronchi into the bronchus, then into the trachea, and then into the outer atmosphere. The process of *breathing the air out of the lungs* is called **EXPIRATION**. When we breathe out, the muscles of the chest make the lungs *contract*.

The respiratory system is important in the production of speech because we need air for speaking. For the production of **MOST SPEECH SOUNDS OF MOST LANGUAGES IN THE WORLD**, we use the air that we **BREATHE OUT**. We can say that we use the **EXPIRATORY LUNG - AIR** for the production of a vast majority of speech sounds used by the languages of the world. In fact, **ALL THE SPEECH SOUNDS (EXCEPT OF ENGLISH AND SEVERAL INDIAN LANGUAGES (EXCEPT Sindhi) ARE PRODUCED WITH THE HELP OF THE AIR THAT WE BREATHE OUT.**

Given on the previous page is a diagram of the organs that comprise the respiratory system. The heart is included in the diagram.

(B) THE ARTICULATORY SYSTEM

The Articulatory System consists of a few organs in our **HEAD** and **NECK**. These organs are called the **ORGANS OF SPEECH**. These organs include the **LIPS**, the **TEETH**, the **TONGUE**, and the **PALATE**. Given below is a diagram of the Organs of Speech.

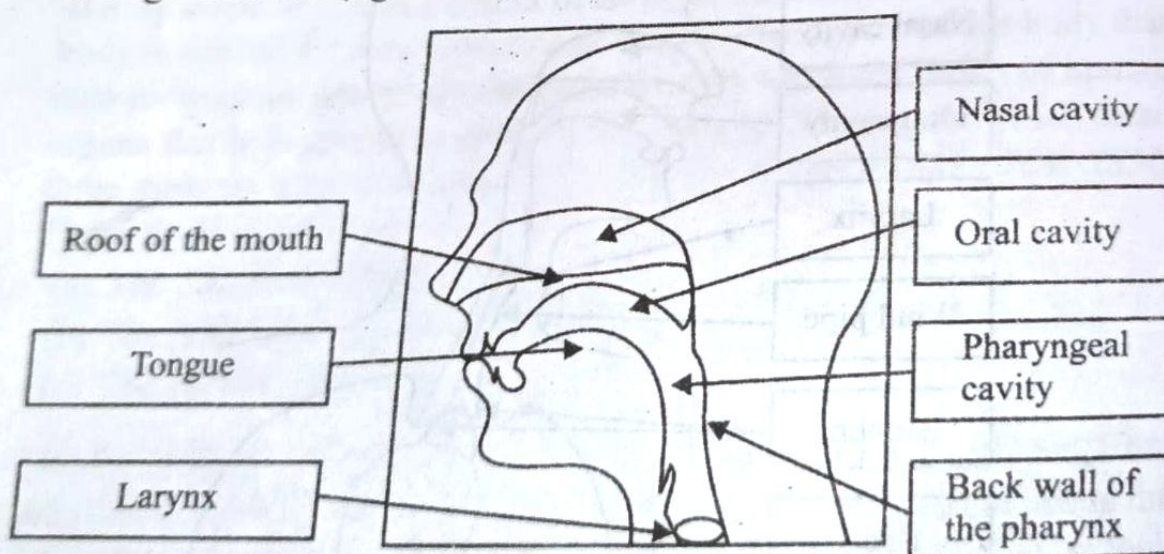


Fig. 4.2 The Organs of Speech

In Fig. 4.2, the nostrils (in fact there are **TWO** nostrils, only one of which is shown in the diagram), the lips, the teeth and the tongue are easily

recognisable. The part called the **ROOF OF THE MOUTH** is divided into four parts and the tongue is divided into six parts. The different parts of the roof of the mouth and the tongue are given in Fig. 4.3.

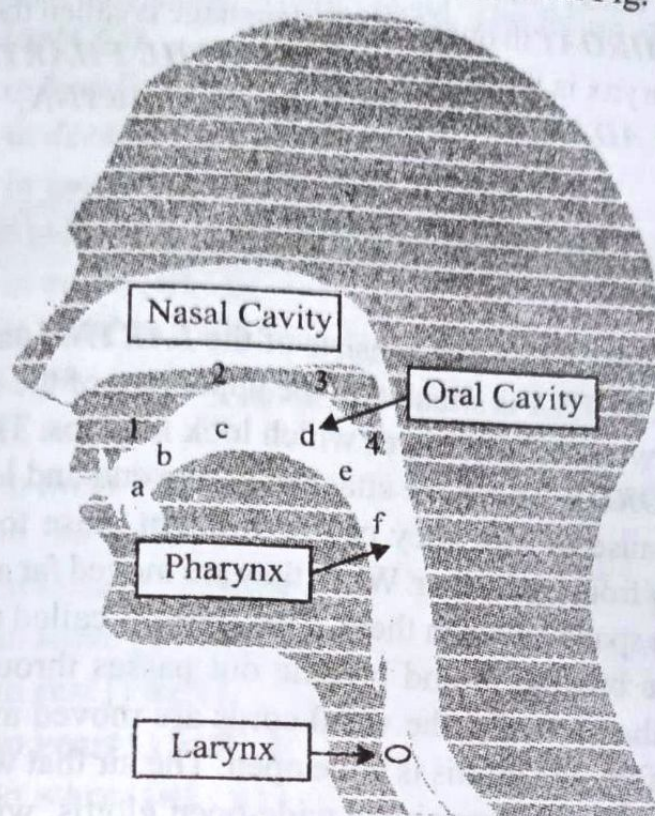


Fig. 4.3 The different parts of the Roof of the Mouth and the Tongue.

Roof of the Mouth: 1. The Teeth Ridge 2. The Hard Palate
3. The Soft Palate 4. The Uvula

Immediately after the front teeth (in the upper row), there is a hard surface on the roof of the mouth. This is called the **TEETH RIDGE** or the **ALVEOLAR RIDGE** or the **ALVEOLUM**. After the teeth ridge the roof of the mouth changes its shape, but continues to be hard and bony. This part of the roof of the mouth is called the **HARD PALATE**. After this, the roof of the mouth becomes soft and fleshy. This part of the roof of the mouth is called the **SOFT PALATE** or the **VELUM**. After the soft palate there is a soft, fleshy part hanging loose. This is called the **UVULA**. The tongue is divided into the **tip**, the **blade**, the **front**, the **centre** and the **back**. The tip can be easily recognised. The **BLADE** of the tongue lies opposite the **TEETH RIDGE**; the **FRONT** lies opposite the **HARD PALATE**; the **BACK** of the tongue lies opposite the **SOFT PALATE**. The **CENTRE** of the tongue lies *between the FRONT and the BACK*. The

ROOT of the tongue lies beyond the back of the tongue. These are the positions occupied by the different parts of the tongue *when the speech organs are AT REST*, that is, when we are not speaking.

What we call **THROAT** in our everyday language is called the **PHARYNX**. Behind the pharynx is the **BACK WALL OF THE PHARYNX**. What we commonly call **ADAM'S APPLE** is called the **LARYNX**.

(C) THE PHONATORY SYSTEM

The **PHONATORY SYSTEM** consists of the **LARYNX** and the **VOCAL CORDS**. The **LARYNX** is situated in the upper part of the trachea. Inside the larynx are two small structures which look like lips. These are called the **VOCAL CORDS**. These are attached at one end and left loose at the other end. Because of this, they can be brought close to each other or moved far away from each other. When they are moved far away from each other, there is a space between them. This space is called the **GLOTTIS**. The air that we breathe in and breathe out passes through the glottis. When we breathe normally, the vocal cords are moved away from each other and therefore, the glottis is wide open. The air that we breathe in or breathe out goes freely through the wide-open glottis, without affecting or moving the vocal cords. When we produce certain speech sounds, the vocal cords are drawn far away from each other and the air passes through the wide-open glottis, without affecting the vocal cords. Such sounds are called **VOICELESS SOUNDS**. In English, the following sounds are **VOICELESS**.

1. [p] as in *spoon* [spu:n]
2. [t] as in *stool* [stu:l]
3. [k] as in *school* [sku:l]
4. [tʃ] as in *cheap* [tʃi:p]
5. [f] as in *feel* [fi:l]
6. [θ] as in *three* [θri:]
7. [s] as in *scene* [si:n]
8. [ʃ] as in *sheet* [ʃi:t]
9. [h] as in *heat* [hi:t]

When we articulate certain speech sounds, the vocal cords are kept close to each other, loosely. When the air from the lungs passes through them, the pressure of the air makes them *open and close rapidly*. This rapid opening and closing of the vocal cords is called the **VIBRATION** of the

vocal cords. Sounds articulated with the vocal cords vibrating are called **VOICED** sounds.

In English, *all the VOWELS* (we shall learn more about **VOWELS** and **CONSONANTS** in Chapter 5) are voiced. The following consonants of English are **VOICED**.

1. [b] as in *beat* [bi:t]
2. [d] as in *deed* [di:d]
3. [g] as in *goose* [gu:s]
4. [dʒ] as in *June* [dʒu:n]
5. [m] as in *mean* [mi:n]
6. [n] as in *need* [ni:d]
7. [ŋ] as in *finger* ['fɪŋgə]
8. [v] as in *veal* [vi:l]
9. [ð] as in *these* [ði:s]
10. [z] as in *zoo* [zu:]
11. [ʒ] as in *leisure* ['leʒə]
12. [l] as in *leave* [li:v]
13. [r] as in *rest* [rest]
14. [j] as in *yeast* [ji:st]
15. [w] as in *wheat* [wi:t]

Given below are two diagrams of the vocal cords and the glottis. Fig. 4.4 shows the vocal cords far away from each other and the glottis wide open. Fig. 4.5 shows the vocal cords drawn close to each other.

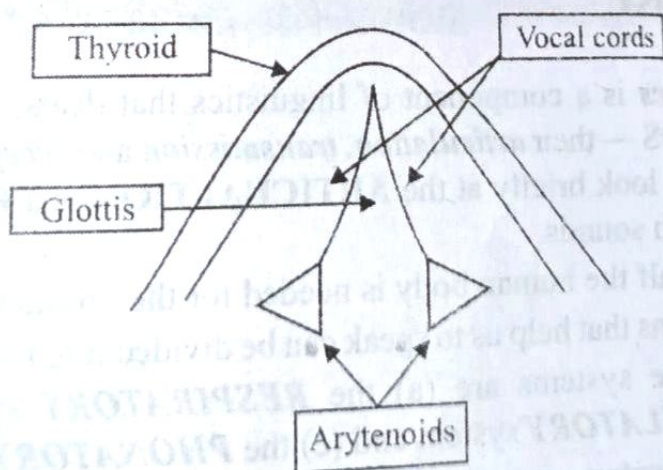


Fig. 4.4 The Vocal Cords and The Glottis
The Vocal Cords are far away from each other.
The glottis is wide open.
This is the position for normal breathing and for
the articulation of **VOICELESS** SOUNDS.

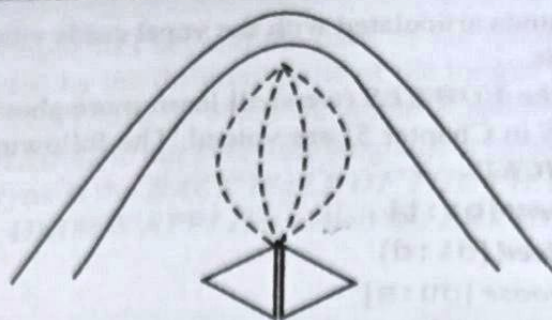


Fig. 4.5 The **VOCAL CORDS AND THE GLOTTIS** — the vocal cords are quite close to each other. This is the position required for the articulation of **VOICED SOUNDS**.

Apart from helping us to articulate voiceless and voiced sounds, the vocal cords perform one more extremely important function. The number of times the vocal cords open and close per second (or **VIBRATE** per second) determines the **PITCH** of our voice. When the vocal cords vibrate **SLOWLY**, our pitch is **LOW**. When they vibrate **FAST**, our pitch is **HIGH**. When we speak normally, we constantly change the pitch of our voice. The way in which we change our pitch when we speak determines the **INTONATION** of a language. We shall discuss intonation in some detail in a later Chapter.

SUMMARY

1. **Phonetics** is a component of linguistics that deals with **SPEECH SOUNDS** — their **articulation**, **transmission** and **reception**. Of these, we shall look briefly at the **ARTICULATION** or **PRODUCTION** of speech sounds.
2. About half the human body is needed for the production of speech.
3. The organs that help us to speak can be divided into three **SYSTEMS**.
4. The three systems are (a) the **RESPIRATORY** system; (b) the **ARTICULATORY** system and (c) the **PHONATORY** system.
5. The **respiratory system** consists of the **lungs**, the **muscles of the chest** and the **windpipe**.
6. Respiration consists of two processes — (a) **inspiration** (or **breathing in**) and (b) **expiration** (or **breathing out**).

7. We use the air that we **breathe OUT** for the production of most speech sounds of the world.
8. The **articulatory system** consists of a few organs in our **head** and **neck**. These organs are called the **ORGANS OF SPEECH**. These organs are the **lips**, the **teeth**, the **tongue** and the **palate**.
9. The **phonatory system** consists of the **larynx** and the **vocal cords**.
10. The vocal cords are situated in the **larynx**. They can be moved close to each other or away from each other. The space between the vocal cords is called the **glottis**.
11. During normal breathing, the vocal cords move far away from each other and therefore the glottis is wide open.
12. The same thing happens when we articulate certain speech sounds. These sounds are called **VOICELESS SOUNDS**.
13. When we articulate certain speech sounds, the vocal cords are drawn close to each other. When the air from the lungs passes through them, the air makes them **VIBRATE**. Speech sounds during the articulation of which the vocal cords vibrate are called **VOICED SOUNDS**.
14. All the **vowels** of English are **VOICED**.
15. **Nine** out of the twenty-four consonants in **English** are **VOICELESS**.
16. **Fifteen** out of the twenty-four consonants of **English** are **VOICED**.
17. The vocal cords perform yet another very important function in speech. The rate at which they vibrate determines the **PITCH** of our voice.
18. When the vocal cords vibrate **SLOWLY**, our pitch is **LOW** and when they vibrate **RAPIDLY**, our pitch is **HIGH**.